

10. For a periodic signal $f(t) = 30 \sin 100t + 10 \cos 300t + 6 \sin(500t + \frac{\pi}{4})$, the fundamental angular frequency in rad/s is ...

- a. 100 b. 150 c. 500 **d. 300**

11. The function $x(t) = \sin(4\pi t) + \sin(5t)$ is signal.

- a. Periodic signal with period $5\pi/2$ **b. Non-periodic** c. Periodic signal with period $1/2$ d. None of them

12. The correct answer that defines the relation between $\text{sgn}(t)$ and $u(t)$ is

- a. $\text{sgn}(t) = u(t) - 1$** b. $\text{sgn}(t) = 2u(t) - 1$ c. $\text{sgn}(t) = 2u(t) + 1$ d. $\text{sgn}(t) = u(t) + 1$

13. The ROC of the Laplace's transform of $x(t) = e^{-t/2} u(-t)$ is

- a. $\text{Re}\{s\} > -2$ **b. $\text{Re}\{s\} < -\frac{1}{2}$** c. $\text{Re}\{s\} < 2$ d. $\text{Re}\{s\} > \frac{1}{2}$ e. $-\frac{1}{2} < \text{Re}\{s\} < 2$

14. The function that does NOT represent a ramp function is...

- a. $r(t) = t$ when $t \geq 0$ b. $r(t) = 0$ when $t < 0$ **c. $r(t) = \int u(t) dt$ when $t < 0$** **d. $r(t) = du(t) / dt$**

15. The system $y(t) = x(2t)$ is

- a. Causal b. Memory **c. Time varying** d. a and b e. a and c **f. b and c**

16. The Laplace transform of the signal $x(t) = te^{-2|t|}$ is

- a. $-\frac{1}{(s-2)^2} + \frac{1}{(s+2)^2}$ **b. $\frac{1}{(s-2)^2} + \frac{1}{(s+2)^2}$** c. $\frac{1}{(s-2)^2} - \frac{1}{(s+2)^2}$ d. $-\frac{1}{(s-2)^2} - \frac{1}{(s+2)^2}$

17. The inverse Fourier transform of $X(j\omega) = 2\pi\delta(\omega) + \pi\delta(\omega - 4\pi) + \pi\delta(\omega + 4\pi)$ is

- a. $x(t) = 1 + \cos(2\pi t)$** b. $x(t) = 1 + \cos(4\pi t)$ c. $x(t) = 1 + 2\sin(2\pi t)$ d. $x(t) = 1 + \cos(2\pi t)$

18. The Fourier transform of $f(x) = \frac{1}{2} \text{rect}(3t/2)$ is

- a. $\text{Sinc}(3\omega/2)$ b. $3 \text{Sinc}(6\omega)$ c. $\frac{1}{2} \text{rect}(6\omega)$ **d. $\frac{1}{3} \text{sinc}(\omega/3)$** e. None of them

***For the given periodic signal $x(t) = 4 + 5 \cos(2\pi 10^6 t - \frac{\pi}{4})$, calculate and plot the Fourier series coefficients to answer the following.

19. The number of coefficients of the Fourier series expansion of the given $x(t)$ is

- a. 1** b. 2 **c. 3** d. 4 e. 5 f. None of them

20. The value of the coefficient C_1 is

- a. 5 b. -5 c. -2.5 d. 4 **e. 2.5**

21. The phases of the coefficients C_0 and C_1 are, respectively.

- a. 0° for both b. $-\frac{\pi}{4}$ and 0° c. 0° and $\frac{\pi}{4}$ **d. 0° and $-\frac{\pi}{4}$** e. $\frac{\pi}{4}$ for both e. None of them

22. The phase spectrum of the coefficient shows

- a. Odd symmetry** b. Even symmetry c. None of them.

$\approx -2\pi j[\delta(w-20\pi) - \delta(w+20\pi)]$, then $f(t)$ is expressed as:
 $\pi \cos(wt)$
 $= 2\pi \sin(wt)$
 $= 2\pi \sin(w-20\pi t)$
 $f(t) = 2 \sin(20\pi t)$

The z-transform of $f[n] = a^n u[n]$ is _____

a. $\frac{1}{z-a}$ b. $\frac{z}{(z-a)}$ c. $\frac{z}{(z-1)^2}$ d. $\frac{z}{z-1}$ e. None of them.

12. All causal systems must be a _____

a. Time invariance b. Memory c. Linearity d. Both b and c e. Both b and d f. Not mentioned.

13. The Laplace Transform of $x(t) = \delta(t+7)$ is given by:

a. e^{7s} b. $\frac{1}{e^{-7s}}$ c. 7 d. πe^{-7s} e. None of them.

14. The Fourier transform of $\text{sinc}\left(\frac{t}{2}\right) * \delta(t)$ is:

a. $\text{sinc}\left(\frac{w}{2}\right)$ b. $\text{sinc}^2\left(\frac{w}{2}\right)$ c. $\text{rect}\left(\frac{w}{2}\right) e^{-\frac{j\pi}{2}}$ d. $2\pi \text{rect}(w)$

15. Which of the following functions can be represented by a Fourier series?

- a. $x(t) = 3 \sin(5t) + \cos\left(3t + \frac{\pi}{4}\right) + 2$
 b. $x(t) = \sin(5t) + \sin(3\pi t)$
 c. $x(t) = t e^{-2t}$
 d. $x(t) = \cos(2t) - 4 \cos\left(\frac{t}{2}\right) + 2$

16. If $x_1(t) = \sin(6t)$ and $x_2(t) = \cos(2\pi t)$ then the signal $x(t) = x_1(t) + x_2(t)$ is described as:

- a. Periodic signal with fundamental period 2π .
 b. Non periodic signal.
 c. Even signal.
 d. Periodic signal with fundamental period $\frac{\pi}{2}$.

17. The Fourier coefficients of the exponential form of $f(t) = \cos(t) + 3 \sin(2t)$, are:

- a. $c_1 = 1, c_2 = 2, c_3 = -2, c_4 = 1.5j, c_5 = -1.5j$
 b. $c_1 = 0, c_2 = \frac{1}{2}, c_3 = \frac{3}{2}, c_4 = -1.5j, c_5 = 1.5j$
 c. $c_1 = 0, c_2 = \frac{1}{2}j, c_3 = -\frac{1}{2}j, c_4 = -1.5, c_5 = 1.5$
 d. $c_1 = 0, c_2 = \frac{1}{2}j, c_3 = -\frac{1}{2}j, c_4 = -2, c_5 = 1$

18. The power of the signal $y[n] = 4$ is _____

- a. 4 b. 12 c. 54 d. 16 e. 8

Q1. Select the correct answer for the following questions:

1. The input-output relationship for LTI system $y(t) = \int_{-\infty}^t 2x(\tau-3)d\tau$, then the impulse response of the system is:

- a. $y(t) = 2$ b. $h(t) = 2\delta(t-3)$ c. $h(t) = 2u(t-3)$ d. $h(t) = 2t$ e. $y(t) = 0$

2. Which of the following system is memoryless?

- a) $h(t) = 0, t \neq 0$. b) $h(t) = x(t-1)$. c) $h(t) = 0, t = 0$. d) $h(t) = kx(t+2)$

3. The Fourier Transform of the signal $x(5t-3)$ in terms of $X(j\omega)$ is given as:

- a. $\frac{1}{5}e^{-j\frac{3\omega}{5}}X(\frac{j\omega}{5})$ b. $\frac{1}{5}e^{\frac{j3\omega}{5}}X(\frac{j\omega}{5})$ c. $\frac{1}{5}e^{-j3\omega}X(\frac{j\omega}{5})$ d. $\frac{1}{5}e^{j3\omega}X(\frac{j\omega}{5})$

4. The impulse response $h(t)$ of linear time-invariant continuous time system is given by $h(t) = e^{-2t}u(t)$, then $H(\omega)$ equal

- a. $\frac{1}{1+2j\omega}$ b. $\frac{\sin \omega}{\omega}$ c. $\frac{1}{2+j\omega}$ d. $\frac{j\omega}{2+j\omega}$

5. For a LTI system for an impulse response $h(t) = e^{2t}u(t-1)$ is considered as....

- a. Causal. b. Memoryless. c. Stable. d. a and b. e. b and c f. Not mentioned.

6. Check the periodicity of the function $x(t) = \cos(2t) + \sin(3t)$. If it is periodic, the fundamental period is....

- a. The function is not periodic.
b. The function is periodic with a fundamental period of 5.2π
c. The function is periodic with a fundamental period of $\frac{5}{4}\pi$
d. The function is periodic with a fundamental period of 2π
e. d. The function is periodic with a fundamental period of $\frac{4}{5}\pi$

7. The given LTI system $y(t) = e^{-4t}u(t)$ is described as:

- a. Unstable and memoryless system.
b. Unstable and causal system.
c. Stable and causal system.
d. Memoryless and causal system.
e. Memoryless, causal and unstable system.

8. The Fourier Transform of $x(t) = \frac{d \operatorname{sgn}(t)}{dt}$ is given by:

- a. $X(\omega) = j\omega \delta(\omega)$ b. $X(\omega) = 2$ c. $X(\omega) = \frac{2}{j\omega} + \pi\delta(\omega)$ d. $X(\omega) = \frac{2}{j\omega}$

9. Which of the following systems is time-invariant?

- a. $y(t) = x(\frac{-t}{2})$ b. $y(t) = x(t)(t-2)$ c. $y(t) = x(1-t)$

$= x(t) + x(t-4)$